



Rikin Pithadia

Roll No : 23035010453

B.Sc Hons. - Data Science and Artificial Intelligence

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EDUCATION

Degree/Certificate	Institute/Board	CGPA	Year
B.Sc Hons. in Data Science and AI	Indian Institute of Technology, Guwahati	7.53	2023 - Present
B.Tech in Mechanical Engineering	Government Engineering College, Gandhinagar	8.52	2023 - Present
Minor in Internet of Things (IoT)	Government Engineering College, Gandhinagar	8.00	2024 - Present

EXPERIENCE

Mirae Flux Labs Pvt. Ltd.(CognifyEV)

Aug 2025 - Nov 2025

Artificial Intelligence Intern

Remote

- Implemented and optimized **RAG-based workflows and knowledge-retrieval pipelines**, improving response accuracy and system reliability across multiple internal AI applications.
- Assisted in building **Agentic AI workflows** and contributed to the development of **Python FastAPI backend services**, including API design, integration, and deployment, resulting in smoother model-backend communication.
- Collaborated with engineering teams using **Git/GitHub, code reviews, branching strategies, and issue tracking**, ensuring seamless development workflows and consistent delivery of RAG and agent automation features.

PROJECTS

1. Semantic Document Q&A & Summarization Assistant | LangChain, RAG, OpenAI

July 2025 - Aug 2025

DocuMind - "Ask. Understand. Summarize"

github.com/rikin-2911/DocuMind

- Developed **DocuMind**, an intelligent **Retrieval-Augmented Generation(RAG)**-based document assistant that enables precise **Q&A and summarization** using information retrieval pipeline from large, unstructured PDF files.
- Utilized **LangChain Orchestration, FAISS vector store, HuggingFace All-MiniLM-L6-V2 embeddings and OpenAI LLM** for semantic search, with document chunking for enhanced **context retrieval**, and deployed via a clean frontend using **Streamlit UI** for real-time, context-aware interactions.

2. Natural Image Classification | CNNs, ResNets, PyTorch, Intel Images

May 2025 - June 2025

Image Classification using Deep Learning

github.com/rikin-2911/ResNet9

- Developed a custom **ResNet9 CNN model** using **PyTorch** to classify natural scene images into **6 categories**, trained on 14,000+ images and obtained **90.27% accuracy** on validation set(3,000+ images) using advanced **image augmentation** techniques(**Resize, Random-Crop, Random-Rotation, Normalization**)
- Automated dataset retrieval via **Kaggle API** and deployed the model with a **Streamlit web app** for real-time image classification and can be utilized in **Terrain Assessment Systems**.

3. Generative AI for Text Generation | LSTM, Deepseek-R1, Streamlit

Feb 2025 - March 2025

Hack the Spring - Technoverse, GEC Gandhinagar

github.com/rikin-2911/TextGenAI

- Built a Generative AI system using **custom LSTM model** for story and dialogue generation, and integrated **Deepseek-R1 LLM via Ollama API** for a generalized **chatbot**, ranked in **top 1%** at GEC Gandhinagar Hackathon.
- Deployed both models through an interactive Streamlit web app. Demonstrated **end-to-end development**, from data preparation to UI deployment, highlighting model versatility and user-centric design.

4. IoT-Based Smart Home Energy Prediction | XGBoost, IoT Data

Jan 2025 - Feb 2025

Integration of AI/ML Systems in IoT

github.com/rikin-2911/IoTEnergyPred

- Designed an **ML pipeline** to predict energy usage in IoT-enabled smart homes using **XGBoost**, and achieved **RMSE score of 0.158** on test data.
- Achieved **97.76% accuracy** on test data through **data preprocessing (imputation, scaling, encoding)** and handled class imbalance effectively.
- Enabled **energy pattern insights** via detailed **EDA** and visualizations using **Seaborn and Matplotlib**.

TECHNICAL SKILLS

- Programming:** Linux Scripting*, C, Python*, R*, Java,
- Python Frameworks:** PyTorch*, TensorFlow, Keras
- Python Tools:** Numpy*, Pandas*, Scikit-learn*, Matplotlib
- Generative AI:** LangChain*, RAG*, LLMs, Prompt Engineering*, HuggingFace models*, FAISS*, Chroma
- Machine Learning:** Supervised and Unsupervised Learning, Algorithms*, Feature Engineering
- Deep Learning:** Computer Vision*, NLP*, CNN, RNN, LSTM, Transformers
- Deployment:** FastAPI*, Docker, Streamlit*, Git, GitHub*

* Elementary proficiency

RELEVANT COURSEWORK

- Mathematics:** Probability and Statistics, Calculus, Linear Algebra, Optimization, Statistical Inferencing
- AI/ML:** Machine Learning Fundamentals, Deep Learning, Time Series Analysis, Recommender Systems
- Electrical and Electronics:** Signals and Systems, AI Based Wireless Communication Systems

CERTIFICATIONS AND ACHIEVEMENTS

• Team Leader - Placement Fair Volunteer Section , GEC-Gandhinagar	<i>2025</i>
• LangChain Essentials for Generative AI , LangChain Web	<i>2025</i>
• Trustworthy Generative AI , Coursera	<i>2024</i>
• Machine Learning and Deep Learning with Python , Jovian	<i>2024</i>
• Business Writing - Effective Communication , Coursera	<i>2024</i>

POSITION OF RESPONSIBILITY

• AI/ML Lead - CLUB IDE (Innovate Develop Engage) , GEC-Gandhinagar
– AI Innovation: Conducted various technical workshops on developing deep learning models with PyTorch and building Agentic AI systems using LangChain and Model Context Protocol (MCP) frameworks. Mentored students in designing modular AI pipelines, tool integration, and real-world deployment strategies.
